

# NGOs, Requirements, and Evaluation Criteria

## Need

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Improving the ease of product retrieval by modifying or adding to the dispenser bin assembly of the original SF vending machine.

## Goals

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1. Reducing the effort needed for product retrieval
2. Adjust the product retrieval height of the vending machine
3. The solution should preserve the integrity of the existing vending machine

| Goals                                                                                | Objective 1                                                                                                                                                                                                                                                              | Objective 2                                                                                                                                                                                                                                                             | Objective 3                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Reducing the effort needed for product retrieval</b>                           | <p>The force required from the user in the complete retrieval process shall not exceed 22N.</p> <p>[CSA/ASC B651.2:25 section 4.2.7]</p>                                                                                                                                 | <p>The depth of the dispenser bin shall not exceed 170 mm. This measurement correlates to average grasp ranges and ensures that products are within a retrievable range.</p> <p>[CSA /ASC B651.2:25 section 9.1.3]</p>                                                  | <p>A product of size 180x100x94 mm shall fit through the opening of the product retrieval container with ease and without causing damage to the product packaging or its contents.</p> <p>[Anthropometric data from 1988 Anthropometric Survey of U.S. Personnel + measurement on products]</p> |
| <b>2. Adjust the product retrieval height of the vending machine</b>                 | <p>Products from the vending machine <b>should</b> be retrieved from a minimum of <b>600 mm</b> above the ground.</p> <p>[Anthropometric data from 1988 Anthropometric Survey of U.S. Personnel + Occupational bending or twisting and low back pain research paper]</p> | <p>Products from the vending machine shall be retrieved from a maximum of 1200 mm from the ground.</p> <p>[CSA/ASC B651 Accessible Design for the Built Environment]</p>                                                                                                | <p>Products from the vending machine shall be retrieved with a reach distance not exceeding 255 mm if there is an obstruction present.</p> <p>[ADA 308.3.2 Accessibility Standards]</p>                                                                                                         |
| <b>3. The solution should preserve the integrity of the existing vending machine</b> | <p>Design shall preserve 100% compliance with all the accessibility standards and operating protocols currently satisfied.</p> <p>[Ambient and Refrigerated Models SERVICE and PARTS MANUAL]</p>                                                                         | <p>Design shall prevent users from bypassing the anti-pilfer system by ensuring no objects with a 12 mm diameter and 328 mm length or larger can infiltrate and dislodge products.</p> <p>[Global Enclosure Standards Enclosure IP ratings as defined by IEC 60529]</p> |                                                                                                                                                                                                                                                                                                 |

| Goals                                                                                | Requirements                                                                                                                                                                       | Evaluation Criteria                                                                                                            |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Reducing the effort needed for product retrieval</b>                           | Force exerted when retrieving the snack (i.e. to open flap, drawer etc.) should not exceed 22N.                                                                                    | The less force required to retrieve the snack, the better.                                                                     |
|                                                                                      |                                                                                                                                                                                    | The shorter the duration of force exertion (the shorter the retrieval process), the better.                                    |
|                                                                                      |                                                                                                                                                                                    | The less body part flexibility required to exert this force, the better.                                                       |
|                                                                                      | The reach depth in order to retrieve the product does not exceed 170 mm in any direction.                                                                                          | The smaller the reach depth, the better.                                                                                       |
|                                                                                      |                                                                                                                                                                                    | The less body part flexibility needed to reach the entire depth, the better.                                                   |
|                                                                                      | Products of size 180 x 100 x 94 mm must be able to be retrieved with ease and without any damage to the product.                                                                   | The less compression or tension the product has to undergo, the better.                                                        |
|                                                                                      |                                                                                                                                                                                    | The more natural and ergonomic the hand movement throughout the retrieval process, the better                                  |
| <b>2. Adjust the product retrieval height of the vending machine</b>                 | Products from the vending machine shall be less than 1200 mm above the ground.                                                                                                     | The closer the retrieval height is to $\frac{600\text{ mm} + 1200\text{ mm}}{2} = 900\text{ mm}$ above the ground, the better. |
|                                                                                      | Products from the vending machine shall be retrieved with a reach distance not exceeding 255 mm if there is an obstruction present.                                                | The closer the obstruction is to 0 mm, the better.                                                                             |
| <b>3. The solution should preserve the integrity of the existing vending machine</b> | The machine as a whole must continue to effectively operate and remain compliant with the 1 accessibility standards and 2 operating protocols currently satisfied.                 | Solutions that help the machine reach additional beneficial standards or better achieve currently compliable ones, the better. |
|                                                                                      |                                                                                                                                                                                    | The less major alterations to the machine, the better.                                                                         |
|                                                                                      | An anti-theft system must be implemented such that no objects with a diameter larger than 12mm and length of 328mm can enter the storage area of the machine to dislodge a product | The prevention of foreign objects that are smaller, the better.                                                                |

## Rescoping (Scoping down)

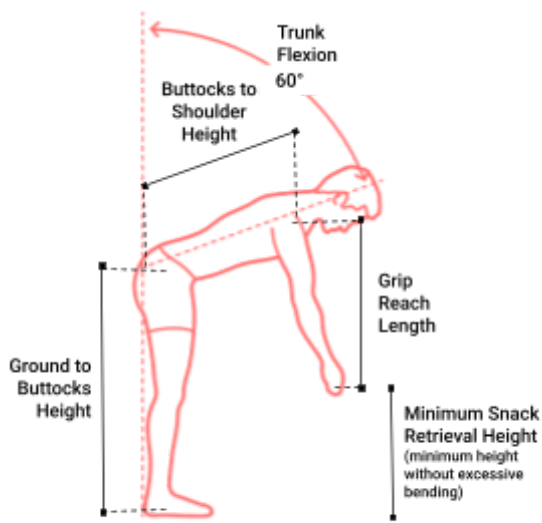
Narrow down from improving ease of product retrieval generally to only modifying / adding to the dispenser bin assembly of the original SF vending machine.

## Reframing

### Goal 1 - General goal

| Change                                                                                                                   | Reason for Change                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| “Reducing the effort needed to <b>reach into the flap</b> ” → “Reducing the effort needed <b>for product retrieval</b> ” | Referring to the flap in the goal was too narrow, as we rescoped to encompass the dispenser bin in general. |

### Goal 2 - Objective 1

| Change                                           | Reason for Change                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Changed “ <b>shall</b> ” to “ <b>should</b> ”    | Bending down (excessive trunk flexion) is not the major concern with the vending machine for the primary stakeholders.<br><br>So it is just something recommended, not mandatory for improving user experience.                                                                                                                                                                                          |
| Changed “ <b>833.1 mm</b> ” to “ <b>600 mm</b> ” | Since excessive trunk flexion is not the major concern, the allowed trunk flexion for this objective is changed from “45°” to “60°”. 60° is the maximum allowable trunk flexion according to ISO 11226:2000 section 3.4.<br> <p><i>Relationship between body metrics and the minimum snack retrieval height.</i></p> |